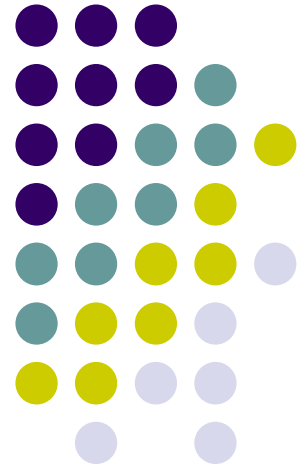
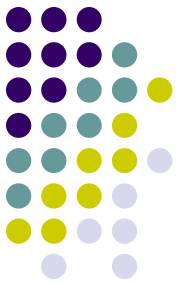


Flutter – Introduction

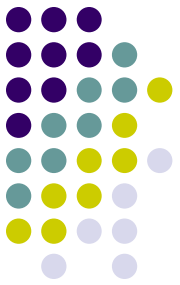
Chapter 01





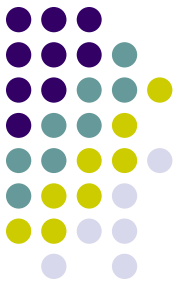
Flutter – Introduction

- In general, developing a mobile application is a complex and challenging task. There are many frameworks available to develop a mobile application. Android provides a native framework based on Java language and iOS provides a native framework based on Objective-C / Swift language.
- However, to develop an application supporting both the OSs, we need to code in two different languages using two different frameworks. To help overcome this complexity, there exists mobile frameworks supporting both OS.
- These frameworks range from simple HTML based hybrid mobile application framework (which uses HTML for User Interface and JavaScript for application logic) to complex language specific framework (which do the heavy lifting of converting code to native code). Irrespective of their simplicity or complexity, these frameworks always have many disadvantages, one of the main drawback being their slow performance.



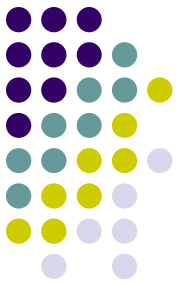
Flutter – Introduction

- In this scenario, Flutter – a simple and high performance framework based on Dart language, provides high performance by rendering the UI directly in the operating system's canvas rather than through native framework.
- Flutter also offers many ready to use widgets (UI) to create a modern application. These widgets are optimized for mobile environment and designing the application using widgets is as simple as designing HTML.
- To be specific, Flutter application is itself a widget. Flutter widgets also supports animations and gestures. The application logic is based on reactive programming. Widget may optionally have a state. By changing the state of the widget, Flutter will automatically (reactive programming) compare the widget's state (old and new) and render the widget with only the necessary changes instead of re-rendering the whole widget.



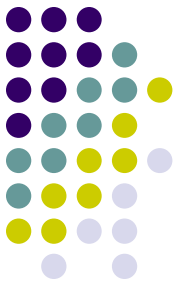
Features of Flutter

- Flutter framework offers the following features to developers:
 - Modern and reactive framework.
 - Uses Dart programming language and it is very easy to learn.
 - Fast development.
 - Beautiful and fluid user interfaces.
 - Huge widget catalog.
 - Runs same UI for multiple platforms.
 - High performance application.



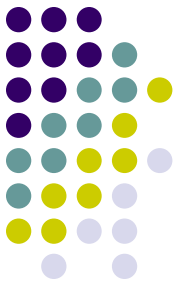
Advantages of Flutter

- Flutter comes with beautiful and customizable widgets for high performance and outstanding mobile application. It fulfills all the custom needs and requirements. Besides these, Flutter offers many more advantages as mentioned below:



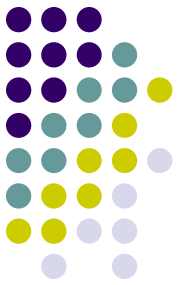
Advantages of Flutter

- Dart has a large repository of software packages which lets you to extend the capabilities of your application.
- Developers need to write just a single code base for both applications (both Android and iOS platforms). *Flutter* may to be extended to other platform as well in the future.
- Flutter needs lesser testing. Because of its single code base, it is sufficient if we write automated tests once for both the platforms.
- Flutter's simplicity makes it a good candidate for fast development. Its customization capability and extendibility makes it even more powerful.
- With Flutter, developers has full control over the widgets and its layout.
- Flutter offers great developer tools, with amazing hot reload.



Disadvantages of Flutter

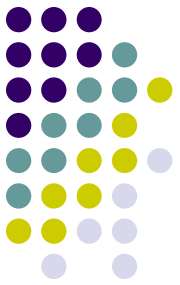
- Despite its many advantages, flutter has the following drawbacks in it:
 - Since it is coded in Dart language, a developer needs to learn new language (though it is easy to learn).
 - Modern framework tries to separate logic and UI as much as possible but, in Flutter, user interface and logic is intermixed. We can overcome this using smart coding and using high level module to separate user interface and logic.
 - Flutter is yet another framework to create mobile application. Developers are having a hard time in choosing the right development tools in hugely populated segment.



Flutter – Installation

- **Installation in Windows:**
 - In this section, let us see how to install *Flutter SDK* and its requirement in a windows system.
 - **Step 1:** Go to URL, <https://flutter.dev/docs/get-started/install/windows> and download the latest *Flutter SDK*. As of April 2019, the version is 1.2.1 and the file is `flutter_windows_v1.2.1-stable.zip`.
 - **Step 2:** Unzip the zip archive in a folder, say `C:\flutter\`
 - **Step 3:** Update the system path to include flutter bin directory.
 - **Step 4:** Flutter provides a tool, flutter doctor to check that all the requirement of flutter development is met.
`flutter doctor`

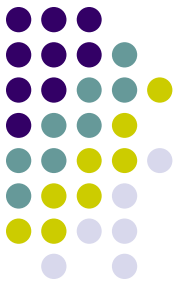
Flutter – Installation



- **Installation in Windows:**

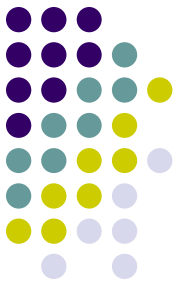
- **Step 5:** Running the above command will analyze the system and show its report as shown below:
- Doctor summary (to see all details, run `flutter doctor -v`): [✓]
Flutter (Channel stable, v1.2.1, on Microsoft Windows [Version 10.0.17134.706], locale en-US) [✓] Android toolchain - develop for Android devices (Android SDK version 28.0.3) [✓] Android Studio (version 3.2) [✓] VS Code, 64-bit edition (version 1.29.1) [!] Connected device ! No devices available ! Doctor found issues in 1 category.
- The report says that all development tools are available but the device is not connected. We can fix this by connecting an android device through USB or starting an android emulator.

Flutter – Installation



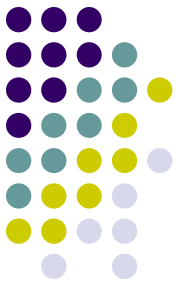
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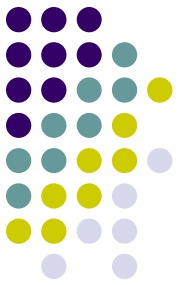
Flutter – Installation

- **Installation in Windows:**
 - **Step 6:** Install the latest Android SDK, if reported by flutter doctor
 - **Step 7:** Install the latest Android Studio, if reported by flutter doctor
 - **Step 8:** Start an android emulator or connect a real android device to the system.
 - **Step 9:** Install Flutter and Dart plugin for Android Studio. It provides startup template to create new Flutter application, an option to run and debug Flutter application in the Android studio itself, etc.,
 - Open Android Studio.
 - Click File > Settings > Plugins.
 - Select the Flutter plugin and click Install.
 - Click Yes when prompted to install the Dart plugin.
 - Restart Android studio.



Flutter – Installation

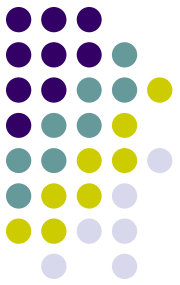
- **Installation in MacOS**
 - To install Flutter on MacOS, you will have to follow the following steps:
 - **Step 1:** Go to URL, <https://flutter.dev/docs/get-started/install/macos> and download latest Flutter SDK. As of April 2019, the version is 1.2.1 and the file is flutter_macos_v1.2.1-stable.zip.
 - **Step 2:** Unzip the zip archive in a folder, say /path/to/flutter
 - **Step 3:** Update the system path to include flutter bin directory (in ~/.bashrc file).
 - > export PATH="\$PATH:/path/to/flutter/bin"



Flutter – Installation

- **Installation in MacOS**
 - **Step 4:** Enable the updated path in the current session using below command and then verify it as well.

```
source ~/.bashrc source $HOME/.bash_profile echo $PATH
```
 - Flutter provides a tool, flutter doctor to check that all the requirement of flutter development is met. It is similar to the Windows counterpart.
 - **Step 5:** Install latest XCode, if reported by flutter doctor
 - **Step 6:** Install latest Android SDK, if reported by flutter doctor
 - **Step 7:** Install latest Android Studio, if reported by flutter doctor
 - **Step 8:** Start an android emulator or connect a real android device to the system to develop android application.



Flutter – Installation

- **Installation in MacOS**
 - **Step 9:** Open iOS simulator or connect a real iPhone device to the system to develop iOS application.
 - **Step 10:** Install Flutter and Dart plugin for Android Studio. It provides the startup template to create a new Flutter application, option to run and debug Flutter application in the Android studio itself, etc.,
 - Open Android Studio.
 - Click **Preferences > Plugins.**
 - Select the Flutter plugin and click Install.
 - Click Yes when prompted to install the Dart plugin.
 - Restart Android studio.

END

